

Prefab Configuration (Runtime Generated)

This project creates all gameplay entities from code at runtime (no manual prefab drag/drop required).

Player “virtual prefab”

Created in `EntityFactory.CreatePlayer` with:

- `SpriteRenderer` (triangle player ship)
- `CircleCollider2D` (`isTrigger = true`)
- `Rigidbody2D` (`isKinematic = true` , `gravityScale = 0`)
- `PlayerController`
- Child `Muzzle` transform
- Child `ShieldVisual` object + `SpriteRenderer`

Enemy “virtual prefab”

Created in `EntityFactory.CreateEnemy` :

- `SpriteRenderer` (color varies by enemy type)
- `CircleCollider2D` trigger
- `Rigidbody2D` kinematic
- `EnemyController`

Enemy types and defaults:

- Basic: HP 20, score 100, single straight bullet
- ZigZag: HP 15, score 130, aimed bullets
- Tank: HP 45, score 220, 3-way spread bullets

Bullet “virtual prefab”

Created in `EntityFactory.CreateBullet` :

- `SpriteRenderer`
- `BoxCollider2D` trigger
- `Rigidbody2D` kinematic
- `BulletController`

Power-up “virtual prefab”

Created in `EntityFactory.CreatePowerUp` :

- `SpriteRenderer`
- `CircleCollider2D` trigger
- `Rigidbody2D` kinematic
- `PowerUpController`

Types:

- `Health` : +30 HP (clamped to max)
- `RapidFire` : faster fire rate for 6 seconds
- `Shield` : invulnerability for 8 seconds